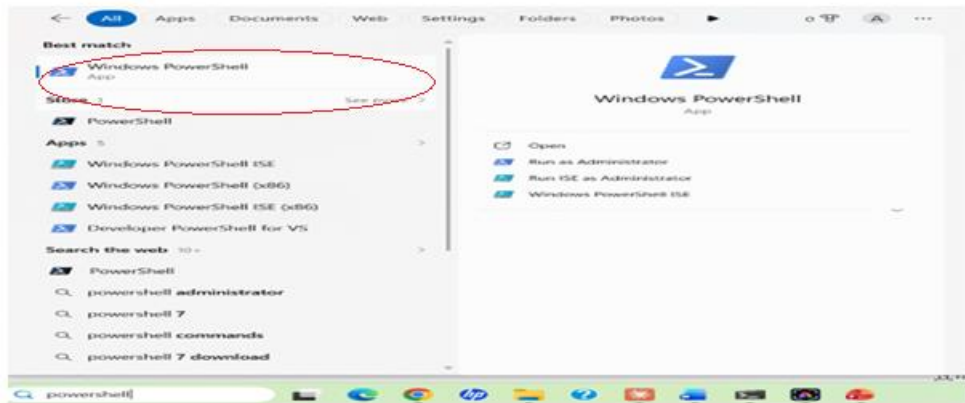
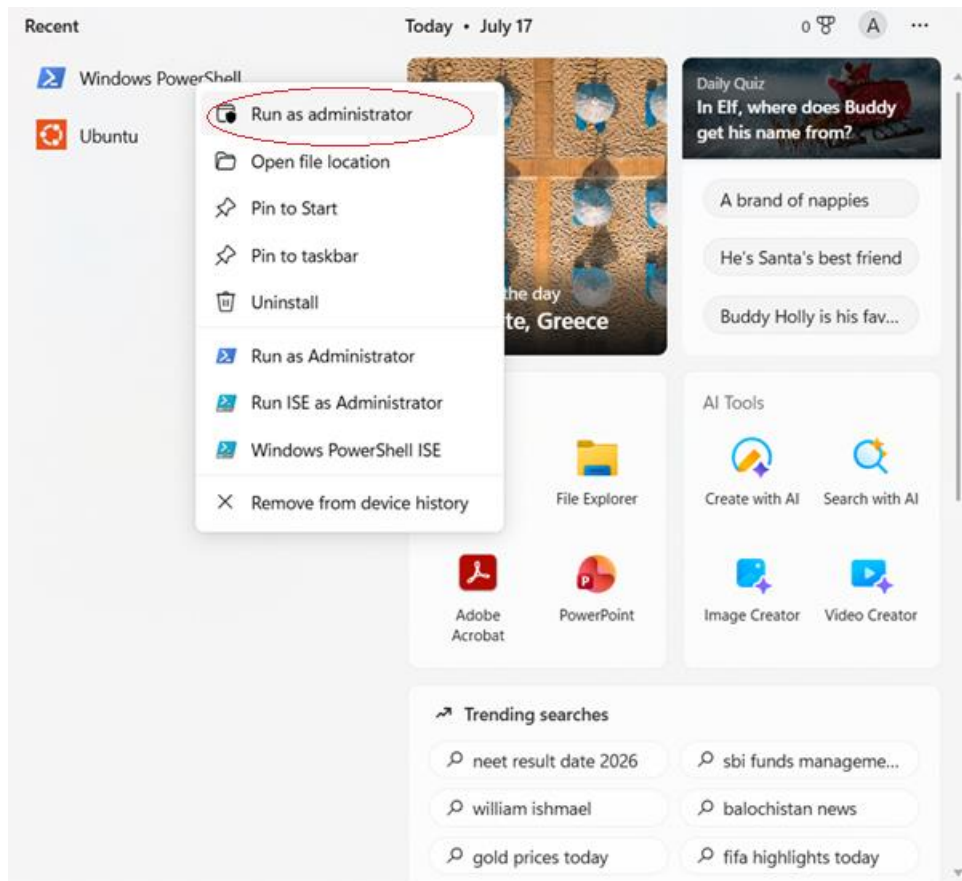


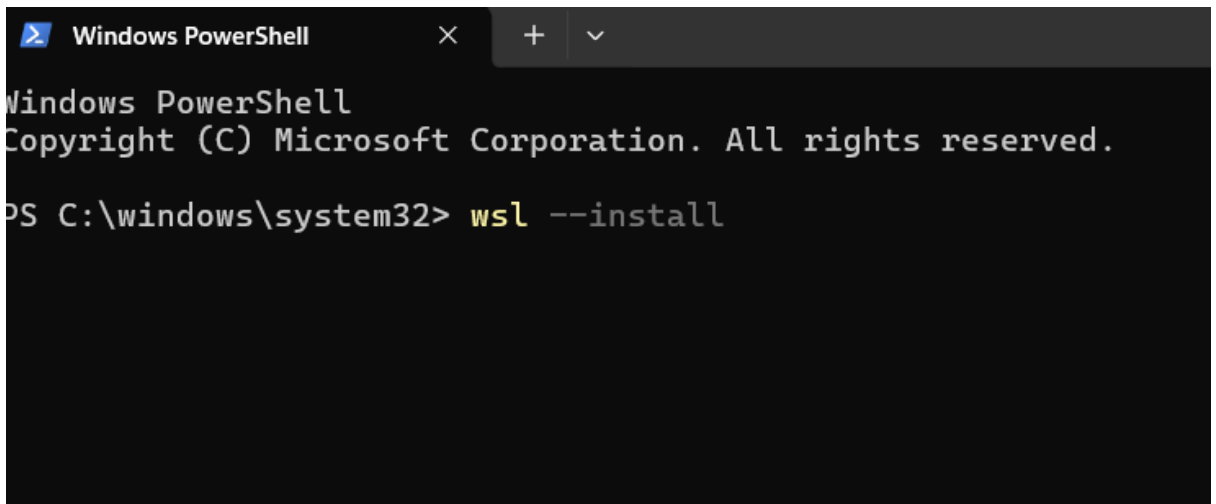
Step 1: Open PowerShell as Administrator

1. Press **Windows**.
2. Search for **PowerShell**.



3. Right-click **Windows PowerShell**.



A screenshot of a Windows PowerShell terminal window. The title bar shows 'Windows PowerShell' with standard window controls. The terminal text reads: 'Windows PowerShell', 'Copyright (C) Microsoft Corporation. All rights reserved.', and 'PS C:\windows\system32> wsl --install'. The 'wsl' command is highlighted in yellow.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\windows\system32> wsl --install
```

Step 3: Restart Your Computer

After installation, restart your PC when prompted.

Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Example:

Enter new UNIX username: anusha

Step 5: Create a Password

You'll then be prompted to create a password.

New password: ****

Retype new password:****

Step 6: Update Ubuntu

```
dijendra@LAPTOP-7C96B10V:~$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [424 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 Packages [1480 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [97.9 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.8 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main amd64 c-n-f Metadata [4616 B]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [159 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [51.2 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [3552 B]
Get:14 http://archive.ubuntu.com/ubuntu resolute/main Translation-en [524 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [361 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute/main amd64 Components [395 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute/main amd64 c-n-f Metadata [32.4 kB]
Get:18 http://security.ubuntu.com/ubuntu resolute/universe amd64 Packages [16.0 MB]
Get:19 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [70.0 kB]
Get:20 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 c-n-f Metadata [396 B]
Get:21 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages [9020 B]
Get:22 http://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [2660 B]
Get:23 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Components [212 B]
Get:24 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 c-n-f Metadata [120 B]
Get:25 http://archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:26 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Components [4556 kB]
Get:27 http://archive.ubuntu.com/ubuntu resolute/universe amd64 c-n-f Metadata [313 kB]
Get:28 http://archive.ubuntu.com/ubuntu resolute/restricted amd64 Packages [152 kB]
```

Step 7: Install GCC (C Compiler)

```
dijendra@LAPTOP-7C96B10V:~$ sudo apt install build-essential
Installing:
  build-essential

Installing dependencies:
  bzip2      g++-15      libalgorithm-diff-xs-perl  libfakeroot      make
  dpkg-dev  g++-15-x86-64-linux-gnu  libalgorithm-merge-perl   libfile-fcntllock-perl
  fakeroot  g++-x86-64-linux-gnu     libcrypt-dev              libstdc++-15-dev
  g++       libalgorithm-diff-perl   libdpkg-perl              lto-disabled-list

Suggested packages:
  bzip2-doc      debian-tag2upload-keyring  g++-15-multilib  bzip      make-doc
  debian-keyring  g++-multilib              gcc-15-doc       libstdc++-15-doc
```

Step 8: Verify the Installation

```
dijendra@LAPTOP-7C96B10V:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

Step 9: Create Your C Program

Example: nano process.c

Step 10: Compile the Program

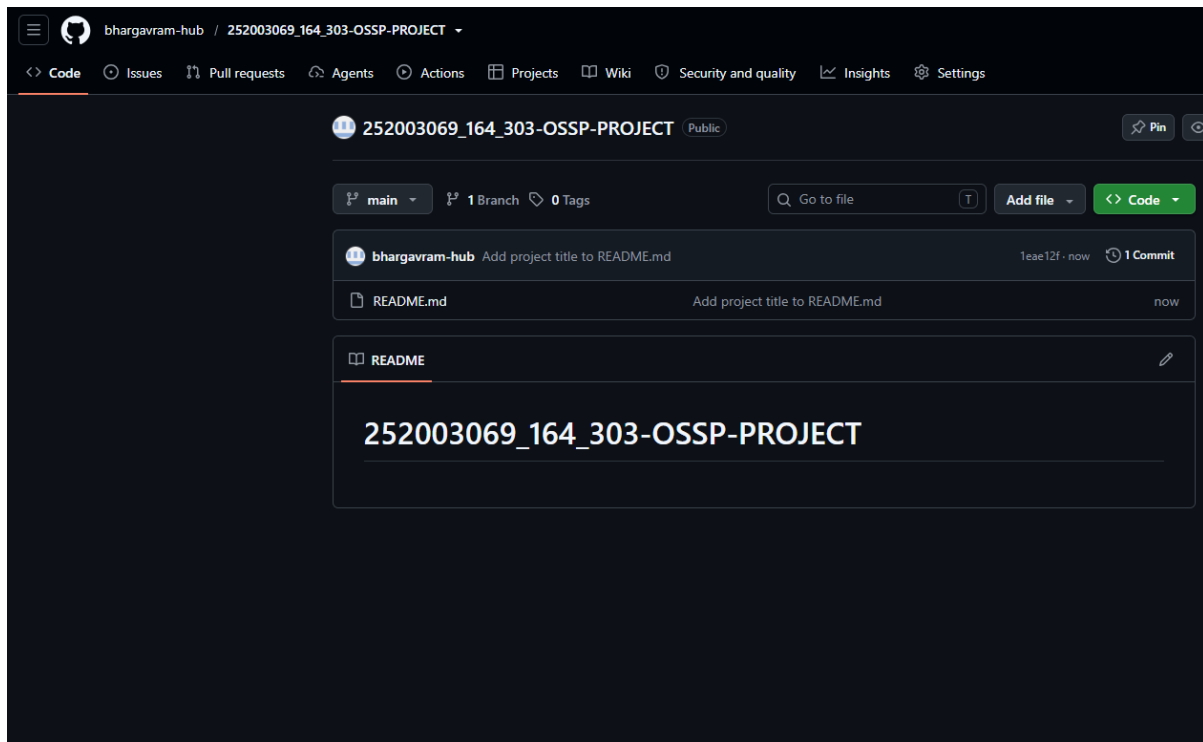
```
dijendra@LAPTOP-7C96B10V:~$ nano hello.c
dijendra@LAPTOP-7C96B10V:~$ gcc hello.c -o hello
dijendra@LAPTOP-7C96B10V:~$ ./hello
Hello Linux
```

4. GITHUB-Repository :

The screenshot shows the GitHub interface for a repository named '2520030164_OSSP' owned by 'dijendramargani07-sudo'. The repository is public and has 1 branch (main) and 0 tags. The file list shows three items: '2520030164_Practical' (added 3 weeks ago), '2520030164_Skill' (added 2 weeks ago), and 'README.md' (initial commit, 3 weeks ago). The README content is visible, showing the title 'OSSP'.

File	Commit Message	Time
2520030164_Practical	Add files via upload	3 weeks ago
2520030164_Skill	Add files via upload	2 weeks ago
README.md	Initial commit	3 weeks ago

5. Project Structure :



6. EXECUTION :

1. FORK() :

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    fork();

    printf("Hello\n");

    return 0;
}
```

Output:

```
dijendra@LAPTOP-7C96B10V:~$ nano fork.c
dijendra@LAPTOP-7C96B10V:~$ gcc fork.c -o fork
dijendra@LAPTOP-7C96B10V:~$ ./fork
Hello
Hello
```

2. EXEC() FAMILY :

```

#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid=fork();

    if(pid==0)
    {
        execlp("date", "date", NULL);
    }
    else
    {
        wait(NULL);

        printf("Child Completed\n");
    }
    return 0;
}

```

Output :

```

Child Completed
diijendra@LAPTOP-7C96B10V:~$ nano fork_exec.c
diijendra@LAPTOP-7C96B10V:~$ gcc fork_exec.c -o fork_exec
diijendra@LAPTOP-7C96B10V:~$ ./fork_exec
Wed Aug 26 03:15:51 UTC 2026
Child Completed

```

3. Parent-Child :

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid;

    pid=fork();

    if(pid==0)
    {
        printf("I am Child Process\n");
    }
    else
    {
        printf("I am Parent Process\n");
    }

    return 0;
}
```

Output :

```
dijendra@LAPTOP-7C96B10V:~$ nano Parentchild.c
dijendra@LAPTOP-7C96B10V:~$ gcc Parentchild.c -o Parentchild
dijendra@LAPTOP-7C96B10V:~$ ./Parentchild
I am Parent Process
I am Child Process
```